LAB EXPERIMENT RECORD

Set 27: Patient Health Risk Analysis

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	N LAHARI
Register Number	192473018
Date	20 October 2026
Day	Tuesday

Dataset: Patient Health Data

Patient ID	Age	BMI	BP	Cholesterol
P1	25	22	120	180
P2	40	28	135	210
P3	55	30	145	240
P4	35	26	130	200
P5	60	32	150	260

1. Scatterplot Matrix

Aim

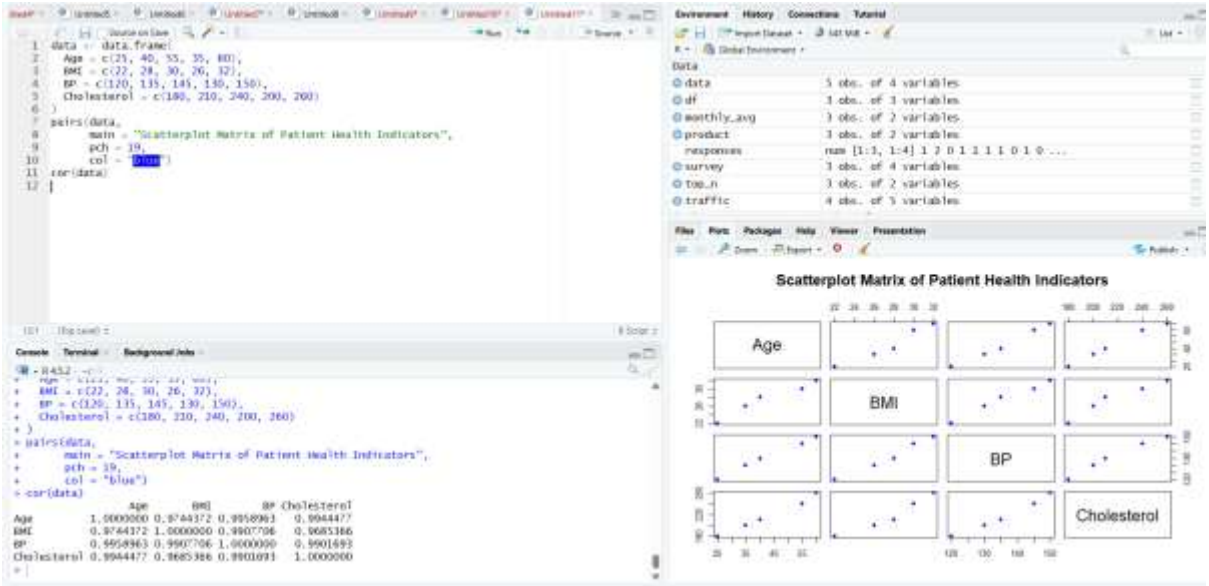
To create a scatterplot matrix for Age, BMI, BP, and Cholesterol to examine pairwise relationships, correlations, and potential health risk patterns.

R Code

R

```
data <- data.frame(
  Age = c(25, 40, 55, 35, 60),
  BMI = c(22, 28, 30, 26, 32),
  BP = c(120, 135, 145, 130, 150),
  Cholesterol = c(180, 210, 240, 200, 260)
)
pairs(data,
      main = "Scatterplot Matrix of Patient Health Indicators",
      pch = 19,
      col = "blue")
cor(data)
```

Output



A scatterplot matrix displaying the relationships between age, BMI, blood pressure, and cholesterol, along with a correlation matrix in the RStudio Console.

Observation

The variables show generally positive relationships in this small dataset. Older patients tend to have higher BMI, blood pressure, and cholesterol values. These patterns suggest possible risk associations but do not establish causation or diagnose a health condition.

2. Q-Q Plot and ECDF for Cholesterol

Aim

To construct a Q-Q plot and an Empirical Cumulative Distribution Function (ECDF) for cholesterol levels to assess distribution behaviour, normality, and skewness.

R Code

R

```
cholesterol <- c(180, 210, 240, 200, 260)

par(mfrow = c(1, 2))

qqnorm(cholesterol,

       main = "Q-Q Plot of Cholesterol",

       pch = 19,

       col = "blue")

qqline(cholesterol, col = "red", lwd = 2)

plot(ecdf(cholesterol),

     main = "ECDF of Cholesterol",

     xlab = "Cholesterol Level",

     ylab = "Cumulative Probability",

     verticals = TRUE,

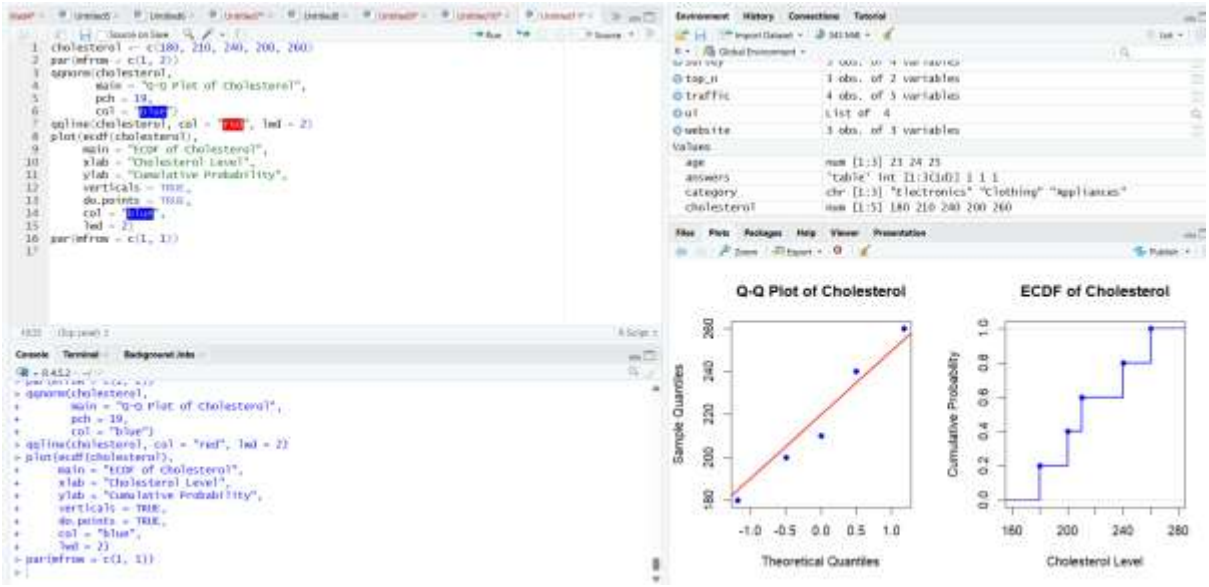
     do.points = TRUE,

     col = "blue",

     lwd = 2)

par(mfrow = c(1, 1))
```

Output



A Q-Q plot comparing cholesterol values with theoretical normal quantiles and an ECDF showing the cumulative distribution of cholesterol levels.

Observation

The Q-Q plot helps assess whether cholesterol values approximately follow a normal distribution. The ECDF increases from 0 to 1 as cholesterol values increase. With only five observations, normality and skewness cannot be assessed reliably.

3. Bar Chart of Average Health Indicators

Aim

To create a bar chart showing the average values of Age, BMI, BP, and Cholesterol with proper axis labels, a descriptive title, and distinct colors.

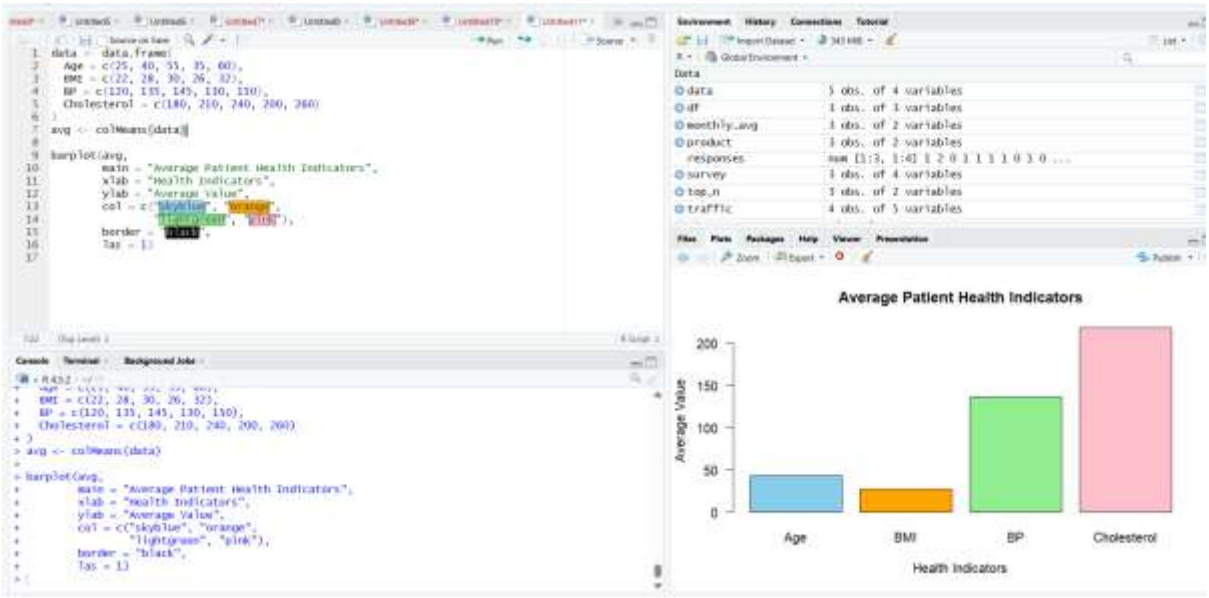
R Code

R

```
data <- data.frame(
  Age = c(25, 40, 55, 35, 60),
  BMI = c(22, 28, 30, 26, 32),
  BP = c(120, 135, 145, 130, 150),
  Cholesterol = c(180, 210, 240, 200, 260)
)
avg <- colMeans(data)
```

```
barplot(avg,
        main = "Average Patient Health Indicators",
        xlab = "Health Indicators",
        ylab = "Average Value",
        col = c("skyblue", "orange",
                "lightgreen", "pink"),
        border = "black",
        las = 1)
```

Output



A bar chart displaying the average values of the four health indicators.

Observation

The average values are:

- Age: 43 years
- BMI: 27.6
- BP: 136
- Cholesterol: 218

These averages summarize the dataset, but the indicators use different units and should not be compared directly by bar height to determine overall risk. Individual clinical interpretation requires appropriate reference ranges and medical context.

4. Interactive Tableau Dashboard

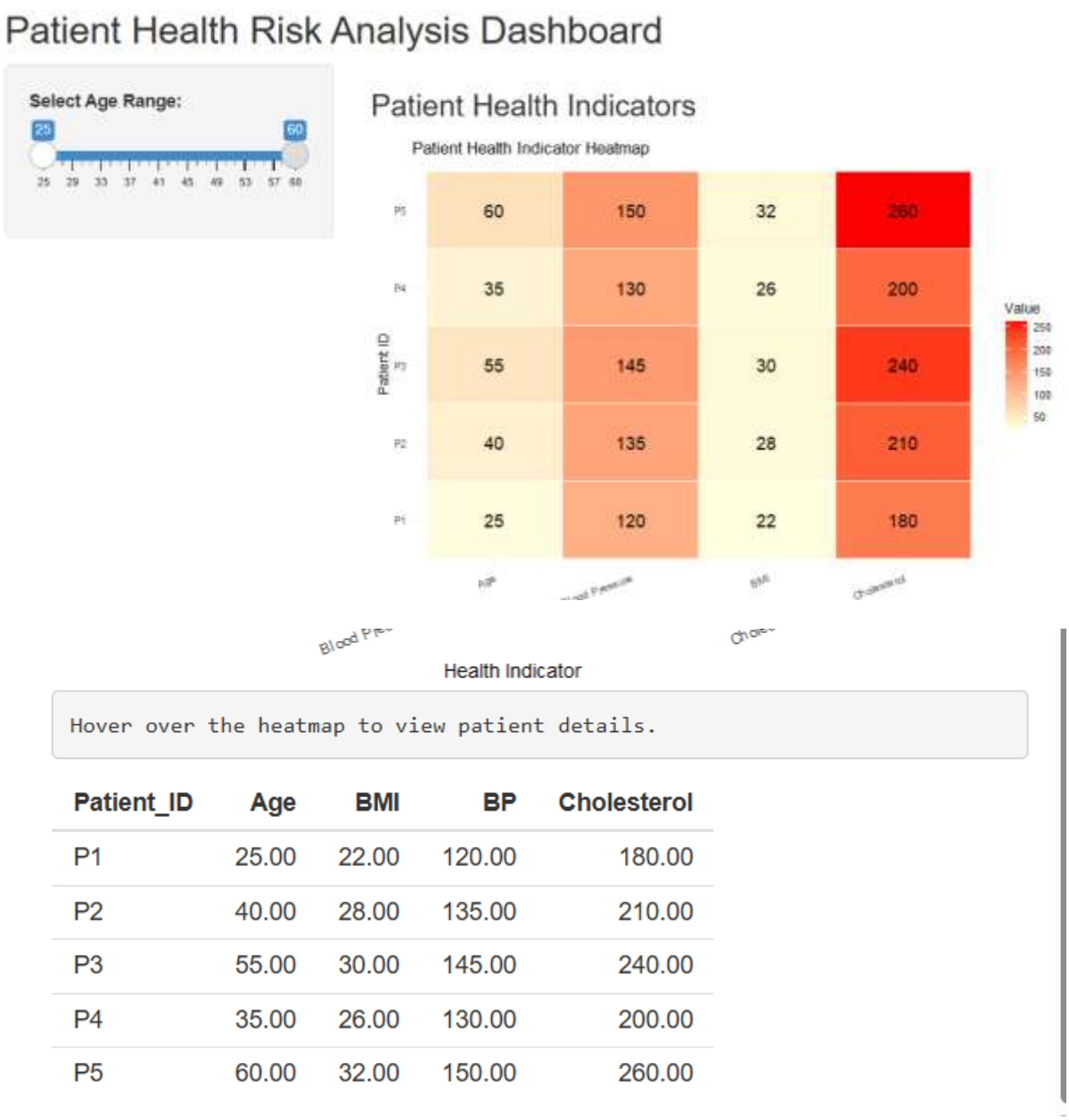
Aim

To create an interactive Tableau dashboard with a heatmap highlighting higher indicator values, an Age filter, and hover tooltips containing all health indicators.

Tableau Procedure

1. Open Tableau and import the patient health dataset.
2. Create a calculated field named **Risk Indicator Score** using a suitable, clearly documented scoring method. The supplied dataset does not include a validated risk classification.
3. Create a worksheet named **Patient Health Heatmap**.
4. Drag Patient_ID to Rows and the health indicators to Columns.
5. Use Measure Values and Measure Names to display the indicator values in a heatmap.
6. Drag Measure Values to Color and select a suitable color gradient.
7. Add Patient_ID, Age, BMI, BP, and Cholesterol to the Tooltip section.
8. Create an Age filter by dragging Age to the Filters shelf and selecting **Show Filter**.
9. Select Dashboard → New Dashboard.
10. Add the heatmap worksheet to the dashboard.
11. Display the Age filter and format the dashboard title as **Patient Health Risk Analysis**.
12. Save the dashboard and test the filter and hover tooltips.

Output



Patient_ID	Age	BMI	BP	Cholesterol
P1	25.00	22.00	120.00	180.00
P2	40.00	28.00	135.00	210.00
P3	55.00	30.00	145.00	240.00
P4	35.00	26.00	130.00	200.00
P5	60.00	32.00	150.00	260.00

An interactive heatmap dashboard that allows users to explore patient health indicators by age and view all recorded measurements through hover tooltips.

Observation

Patient P5 has the highest age, BMI, blood pressure, and cholesterol values in this dataset, while P3 also has relatively high values. These observations can help identify records for further review, but they are not a validated clinical risk ranking. The Age filter allows healthcare professionals to explore patterns across different age groups.